School of Computer Science
University of Southampton Malaysia

Neu You Quan

An Optimised Machine Learning Model for Gallstone Disease Prediction

Project supervisor: Dr. Abobakr Al-Shamiri
Second examiner: Dr. Muhamad Najib Zamri

A project progress report submitted for the award of
Bachelor of Science (BSc) in Computer Science

Contents

Abstract 3
Background 4
Gallstone 4
Machine Learning 4
Literature Review 6
Machine Learning in Healthcare and Hepatobiliary 6
Machine Learning and Gallbladder Disease 6
Project Goal 9
Technical Progress 10
Dataset 10
Data Preprocessing 13
Model Implementation 13
Plan of Remaining Work 14
Reference 15

Abstract

Gallstone disease (GD) is one of the most common disorders affecting the digestive system worldwide, with rising incidence due to a combination of genetic, biological, and lifestyle factors. Traditional diagnostic methods such as ultrasonography, CT scans, and MRI are effective but may be costly and less accurate in certain patient populations. This project aims to design and develop a machine learning-based model, called Extreme Learning Machine (ELM) capable of predicting the risk of gallstone disease using clinical and demographic data. In addition, metaheuristic algorithms such as Artificial Bee Colony (ABC) algorithm will be utilised to optimize the performance of ELM model. Comprehensive documentation of the data preprocessing, feature selection, model training, and optimization stages will be presented in the final report.

Background

Gallstone

Gallstones disease, also known as cholelithiasis, is a prevalent biliary disorder worldwide which generally develop through the progressive deposition of cholesterol on an initial precipitation nidus (Sharma, Sharma and Prasad, 2024). The prevalence is differed to region and country, reported approximately 10-20% in Western countries and 5-12% in Eastern countries at some time (Stinton and Shaffer, 2012). Even though Later reports indicated that approximately 6% of the global population has gallstones since 21st century, some others journal proposed that 50-70% of patient are asymptomatic when diagnosis, while the incidence rate has almost tripled in the past 30 years (Sakorafas, Milingos and Peros, 2007; Unalp-Arida and Ruhl, 2023; Wang *et al.*, 2024; Muraviev *et al.*, 2025). However, it is certain that the situation has not improved, as the mortality rate and social burden caused by gallstones are increasing (Ruhl and Everhart, 2011; Unalp-Arida and Ruhl, 2023, 2024).

Currently, diagnosis of cholelithiasis is primarily conducted through systematically transcutaneous sonography (abdominal ultrasound), which provide 95% of sensitivity and practically 100% of specificity, supporting the high confidence for medical staff (Gutt, Schläfer and Lammert, 2020). After diagnosing, treatment is proceeded while the most preferred method currently is cholecystectomy. Despite this treatment process is formal and effective, but it also comes with disadvantages, such as the patients may have issues with fat digestion as their gallbladder is removed along with the gallstone (Sharma, Sharma and Prasad, 2024). Furthermore, lack of more suitable treatment, high cost of treatment, failure to detect asymptomatic patients as non-active medical check-up, variety kind of complication or postoperative sequelae etc (Götzky, Landwehr and Jähne, 2013; Sakai *et al.*, 2024; Khandelwal, Anand and Kant, 2025) . To optimize these, artificial intelligent has brought forward to the field, boosting medical process from variety part such as early detection of disease, course prognosis and risk estimation etc (Khalifa and Albadawy, 2024).

Machine Learning

Machine learning is a subfield of Artificial Intelligence, with the concept of building a system which can improve its performance measure on certain tasks after experience with specific learning (Mitchell, 1997). Machine learning is built on three key elements: data (the 'experience' for learning), algorithms (the core calculations performed to get results), and models (the combination and formation of a 'black box' that describe the generalized relationship of input variables and output variables) (Mahalle *et al.*, 2025) . It aims to detect and learn autonomously the inner patterns of data through limited training data and learning algorithms, then performs some tasks based on the data never seen before, addressing specify

problems clearly defined by human (Jiang, Ma and Wu, 2026). By applying machine learning, robust support is provided to some complex scenarios, such as those tasks requiring tremendous tuning, tasks performing in fluctuating environment, complex problems lacking established solutions, or extracting insights within enormously dataset (Farnham *et al.*, no date) .

Literature Review

Machine Learning in Healthcare and Hepatobiliary

Machine learning had been utilised widely in healthcare discipline nowadays to support medical professionals' process of diagnosing and treating, and achieve result in diagnosis precision enhancement, administrative procedures simplification and treatment regimens individualization etc (An *et al.*, 2023). Notable results include the conversion of large-scale clinical data into actionable insights, tumour detection and classification in oncology, and treatment guidance and antimicrobial resistance monitoring in clinical microbiology (Waring, Lindvall and Umeton, 2020; Ashraf *et al.*, 2025).

Hepatobiliary department is not excepted; in fact, there are multiple and increasingly study of corporation of machine learning (ML) or artificial intelligence (AI) in hepatobiliary related disease currently. For example, Srivastava et al. attempted three machine learning models based on electronic health record datasets for the prediction and classification of liver diseases (Srivastava *et al.*, 2022). Alternatively, Akabane et al. proposed a model for possibility of Endoscopic Retrograde Cholangiopancreatography (ERCP) prediction after emergency laparoscopic cholecystectomy with medical record data and CT/Ultrasound features (Akabane *et al.*, 2025).

Machine Learning and Gallbladder Disease

As the diagnosis of gallstone disease currently are highly reliant on medical records and medical images, the attempts of improvement through machine learning techniques are clearly appropriate. Ahmed et al. have done a systematic review of accuracy of gallstone diagnosis using machine learning technologies image analysis, the results concluded that most models demonstrated good accuracy and speed (Ahmed *et al.*, 2024). Variety of ML models were approached and proposed over time. For example, Pang et al. built a novel YOLOv3-arch neural network for cholelithiasis identification and gallstones classification while Kumar et al. developed a custom Convolutional Neural Network (CNN) architecture for gallstone classification and prediction (Pang, Ding, *et al.*, 2019; Kumar *et al.*, 2025). However, the robustness of models across different clinical settings and devices are concerned due to limitations such as generalization constraint, limited image types of reliance, external validation lacking, and computational demands of complex models etc (Ahmed *et al.*, 2024).

Researchers have attempted various methods to overcome these limitations. For instance, an AI diagnostic system for cholelithiasis recognition was developed by Pang et al. based on lightweight CNN concepts, which was able to run on mobile devices of Android platform (Pang,

Wang, *et al.*, 2019). A model took point-of-care ultrasound (POCUS) as input for cholelithiasis and cholecystitis detection was proposed by Yu et al. to assist the emergency physicians, even in remote areas (Yu *et al.*, 2021). On the other hand, Kumar et al. improved the model generalization by producing synthetic ultrasound images through a modified deep convolutional generative adversarial network (DCGAN) in addition. At the same time, a hybrid visual explanation method for CNN to explain model prediction was proposed to increase the reliability and explainability (Kumar *et al.*, 2025).

Although various attempts have been made to overcome ML limitations, some challenges inheritance from the ultrasound and CT techniques themselves, such as high-intensity background tissue, non-uniform intensity, and image artifacts. To address these issues, other predictive methods have been employed, such as using clinical laboratory data. Some examples are Salem et al. tried machine learning approaches for gallstone disease prediction during their research on identification of metabolites and clinical features associated with gallstone disease, while Blum demonstrated a ML model which predict choledocholithiasis by using patient's digital medical records (Salem *et al.*, 2023; Blum *et al.*, 2024). A new dataset of bioimpedance and laboratory data was curated and constructed by Esen et al. for early prediction of gallstone disease, and nine ML techniques had been experimented through by them and 85.42% of accuracy was achieved by gradient boosting technique as the best result (Esen *et al.*, 2024). Building upon this, deep learning strategy which optimized Random Forest (RF) classifier by different algorithm for gallstone prediction was developed by Sarker et al. but only achieved approximately 4% lower result by using the same dataset built by the former (Sarker, Tiang and Nahid, 2025).

Discarding gallstone diagnosis, ML technologies also been employed for other outcome predictions in gallstone patients, including major complications of gallstone disease, the difficulty of laparoscopic surgery, and malignancy risk etc (Huang *et al.*, 2024, 2025; Ma *et al.*, 2024; Cetin *et al.*, 2025). The reliability of the models should be confident enough as the techniques affecting wide area, the evaluation and selection of the model come to the front. Along hepatobiliary field, area under the receiver operating characteristic (AUROC) and accuracy combine with cross-validation are the most common evaluation methods used. As hepatobiliary diseases diagnosis was mainly gone through clinical images, CNN outperformed as most common selection and deployed in most computer aided diagnosis system, the models proposed by researchers are also mostly developed on CNN (Ahmed *et al.*, 2024; Kumar *et al.*, 2025). In contrast, variety of ML techniques were used in case of prediction based on laboratory data, regression and support vector machine were showing most often.

Despite of more variance of models were used, the research based on laboratory data is still insufficient compared to research based on images, especially in cholelithiasis prediction. Furthermore, although there were multiple models and algorithms had been attempted, but there are still many of models haven't been utilized. In addition, there was still gap between the evaluation results of laboratory data-based models with images-based models, with the image-based models are generally 5~10% better compared to data-based models. This might reason by insufficient data, and laboratory data fluctuation as affected by overall body status while images-based data only focusing on region of interest. Therefore, there is still a clear need to bridge this performance gap by finding new ways to refine and optimize machine learning models that rely solely on clinical and laboratory data.

Project Goal

This project aims to design and develop a machine learning-based model, called Extreme Learning Machine (ELM) capable of predicting the risk of gallstone disease using clinical and demographic data. The study further seeks to employ metaheuristic algorithms to optimise model performance, thereby improving predictive accuracy and reliability. This project will focus on the development of an optimised ELM for gallstone disease prediction using publicly available medical datasets. The project will include data preprocessing, feature selection, model training, evaluation, and optimisation using metaheuristic algorithms such as Artificial Bee Colony (ABC) algorithm. Excluded from the project are clinical trials, patient data collection from hospitals, and the design of new diagnostic hardware. The deliverables will include the trained models, performance evaluation metrics, comparative results before and after optimisation, and documentation of methodology.

Technical Progress

Dataset

The gallstone disease diagnosis dataset is obtained from UC Irvine Machine Learning Repository (UCI), which collected and provided by (Esen *et al.*, 2024), total 319 instances with 39 features are included, and its information is shown in the Table 1. The 'Gallstone Status' column is the target feature, with 1 representing the true diagnosis of gallstone while 0 represents the healthy status. The remaining 38 feature columns include:

- Demographics: Age, Gender, Height, Weight, Body Mass Index (BMI).
- Comorbidities: Comorbidity, Coronary Artery Disease (CAD), Hypothyroidism, Hyperlipidemia, Diabetes Mellitus (DM).
- Bioimpedance Data : Total Body Water (TBW), Extracellular Water (ECW), Intracellular Water (ICW), Extracellular Fluid/Total Body Water (ECF/TBW), Total Body Fat Ratio (TBFR), Lean Mass (LM), Body Protein Content (Protein), Visceral Fat Rating (VFR), Bone Mass (BM), Muscle Mass (MM), Obesity, Total Fat Content (TFC), Visceral Fat Area (VFA), Visceral Muscle Area (VMA), Hepatic Fat Accumulation (HFA).
- Laboratory Data: Glucose, Total Cholesterol (TC), Low Density Lipoprotein (LDL), High Density Lipoprotein (HDL), Triglyceride, Aspartat Aminotransferaz (AST), Alanin Aminotransferaz (ALT), Alkaline Phosphatase (ALP), Creatinine, Glomerular Filtration Rate (GFR), C-Reactive Protein (CRP), Hemoglobin (HGB), Vitamin D.

The dataset was done processing by the creators, resulting in a complete dataset, there is no missing value needed to be filled in and also all values are already in numerical forms, with 0 or 1 for binary categorical type, numeric for continuous type, hepatic fat accumulation is the only special with 0,1,2,3 for categorical type. The dataset is also a balanced dataset in terms of gallstone status, with 158 true value (68 Male;90 Female) and 161 false values (94 Male; 67 Female). The descriptive statistics for the dataset, including mean, standard deviation, and the five-number summary (minimum, quartiles, and maximum) are presented in Table 2.

Table 1. Dataset Information

		Feature	Type	Description	Value / Unit
Target	1	Gallstone Status	Cat.	Presence of Gallstone	0: No, 1: Yes
Demographic Variables	2	Age	Cont.	Age of the person	Years
	3	Gender	Cat.	Gender of the person	0: Male, 1: Female
	4	Height	Cont.	Barefoot height	cm
	5	Weight	Cont.	Body weight with thin clothing	kg
	6	Body Mass Index (BMI)	Cont.	Weight / Height²	kg/m²
Medical History	7	Comorbidity	Cat.	Concomitant diseases	0: No, 1: Yes
	8	Coronary Artery Disease (CAD)	Cat.	Cardiovascular disease	0: No, 1: Yes
	9	Hypothyroidism	Cat.	Underactive thyroid gland	0: No, 1: Yes
	10	Hyperlipidemia	Cat.	High levels of fat in the blood	0: No, 1: Yes
	11	Diabetes Mellitus (DM)	Cat.	High blood sugar	0: No, 1: Yes
Bioimpedance (BIA)	12	Total Body Water (TBW)	Cont.	Total water in the body	kg
	13	Extracellular Water (ECW)	Cont.	Total Extracellular water	kg
	14	Intracellular Water (ICW)	Cont.	Total Intracellular water	kg
	15	Extracellular Fluid/Total Body Water (ECF/TBW)	Cont.	Edema Index	
	16	Total Body Fat Ratio (TBFR)	Cont.	Total fat content	%
	17	Lean Mass (LM)	Cont.	Lean body mass	%
	18	Body Protein Content (Protein)	Cont.	Protein content of the body	%
	19	Visceral Fat Rating (VFR)	Cont.	Visceral organ fat level	
	20	Bone Mass (BM)	Cont.	Bone weight	kg
	21	Muscle Mass (MM)	Cont.	Amount of muscle	kg
	22	Obesity	Cont.	Degree of obesity	%
	23	Total Fat Content (TFC)	Cont.	Total fat amount	kg
	24	Visceral Fat Area (VFA)	Cont.	Inner adipose tissue area	kg
	25	Visceral Muscle Area (VMA)	Cont.	Inner muscle area	kg
	26	Hepatic Fat Accumulation (HFA)	Cat.	Fatty liver stage	0,1,2,3
Laboratory Data	27	Glucose	Cont.	Blood glucose level	mg/dL
	28	Total Cholesterol (TC)	Cont.	TC level	mg/dL
	29	Low Density Lipoprotein (LDL)	Cont.	LDL level	mg/dL
	30	High Density Lipoprotein (HDL)	Cont.	HDL level	mg/dL
	31	Triglyceride	Cont.	Triglyceride level	mg/dL
	32	Aspartat Aminotransferaz (AST)	Cont.	Liver enzyme level	U/L
	33	Alanin Aminotransferaz (ALT)	Cont.	Liver related enzyme level	U/L
	34	Alkaline Phosphatase (ALP)	Cont.	Liver and bone enzyme level	U/L
	35	Creatinine	Cont.	Kidney function indicator	mg/dL
	36	Glomerular Filtration Rate (GFR)	Cont.	Kidney filtration rate	ml/seconds
	37	C-Reactive Protein (CRP)	Cont.	Inflammation indicator	mg/L
	38	Hemoglobin (HGB)	Cont.	Hemoglobin concentration	g/dL
	39	Vitamin D	Cont.	Vitamin D concentration	ng/mL

Table 2. Statistical Information of Dataset

	count	mean	std	min	25%	50%	75%	max
Gallstone Status	319	0.4953	0.5008	0	0	0	1	1
Age	319	48.0690	12.1146	20	38.5	49	56	96
Gender	319	0.4922	0.5007	0	0	0	1	1
Comorbidity	319	0.3354	0.5173	0	0	0	1	3
Coronary Artery Disease (CAD)	319	0.0376	0.1906	0	0	0	0	1
Hypothyroidism	319	0.0282	0.1658	0	0	0	0	1
Hyperlipidemia	319	0.0251	0.1566	0	0	0	0	1
Diabetes Mellitus (DM)	319	0.1348	0.3420	0	0	0	0	1
Height	319	167.1567	10.0530	145	159.5	168	175	191
Weight	319	80.5649	15.7091	42.9	69.6	78.8	91.25	143.5
Body Mass Index (BMI)	319	28.8771	5.3137	17.4	25.25	28.3	31.85	49.7
Total Body Water (TBW)	319	40.5878	7.9302	13	34.2	39.8	47	66.2
Extracellular Water (ECW)	319	17.0712	3.1619	9	14.8	17.1	19.4	27.8
Intracellular Water (ICW)	319	23.6345	5.3493	13.8	19.3	23	27.55	57.1
Extracellular Fluid/Total Body Water (ECF/TBW)	319	42.2120	3.2445	29.23	40.075	42	44	52
Total Body Fat Ratio (TBFR) (%)	319	28.2750	8.4444	6.3	22.025	27.82	34.81	50.92
Lean Mass (LM) (%)	319	71.6382	8.4376	48.99	65.165	72.11	77.85	93.67
Body Protein Content (Protein) (%)	319	15.9388	2.3347	5.56	14.465	15.87	17.43	24.81
Visceral Fat Rating (VFR)	319	9.0784	4.3325	1	6	9	12	31
Bone Mass (BM)	319	2.8033	0.5095	1.4	2.4	2.8	3.2	4
Muscle Mass (MM)	319	54.2730	10.6038	4.7	45.8	53.9	62.6	78.8
Obesity (%)	319	35.8501	109.7997	0.4	13.9	25.6	41.75	1954
Total Fat Content (TFC)	319	23.4878	9.6076	3.1	17	22.6	28.55	62.5
Visceral Fat Area (VFA)	319	12.1716	5.2622	0.9	8.57	11.59	15.1	41
Visceral Muscle Area (VMA) (Kg)	319	30.4034	4.4605	18.9	27.25	30.40811	33.8	41.1
Hepatic Fat Accumulation (HFA)	319	1.1505	1.0592	0	0	1	2	4
Glucose	319	108.6887	44.8487	69	92	98	109	575
Total Cholesterol (TC)	319	203.4953	45.7585	60	172	198	233	360
Low Density Lipoprotein (LDL)	319	126.6524	38.5412	11	100.5	122	151	293
High Density Lipoprotein (HDL)	319	49.4755	17.7187	25	40	46.5	56	273
Triglyceride	319	144.5022	97.9045	1.39	83	119	172	838
Aspartat Aminotransferaz (AST)	319	21.6850	16.6976	8	15	18	23	195
Alanin Aminotransferaz (ALT)	319	26.8558	27.8844	3	14.25	19	30	372
Alkaline Phosphatase (ALP)	319	73.1125	24.1811	7	58	71	86	197
Creatinine	319	0.8006	0.1764	0.46	0.65	0.79	0.92	1.46
Glomerular Filtration Rate (GFR)	319	100.8189	16.9714	10.6	94.17	104	110.745	132
C-Reactive Protein (CRP)	319	1.8539	4.9896	0	0	0.215	1.615	43.4
Hemoglobin (HGB)	319	14.4182	1.7758	8.5	13.3	14.4	15.7	18.8
Vitamin D	319	21.4014	9.9817	3.5	13.25	22	28.06	53.1

Data Preprocessing

According to the abstract on UCI page of the dataset, additional preprocessing is generally unnecessary. However, data from Table 2. Statistical Information of DatasetTable 2 shows that there are some extreme high standard deviations across the columns, indicating that some columns values are containing conspicuous outliers. Furthermore, there are different data requirements for different machine learning models. Hence, the outliers were handled by capping techniques, which mean the data located outside the boundaries are replaced with the boundary values. Afterward, the data were processed with scaling as ELM is sensitive to this. Data scaling was performed by Min-Max Scaler, which scaled the data into the range of 0 to 1 according to their position over the range in its columns.

Model Implementation

An ELMClassifier was imported from scikit-elm library, and the dataset was partitioned into 80% training and 20% testing sets. By using a straightforward processing pipeline, the model was trained and evaluated without cross-validation, which achieved 0.78 accuracy and 0.78 F1-score without cross validation. However, the primary objective of this project is the development of a custom-built, optimized ELM model, which is still in early stages, and is detailed in plan of remaining works section. More detailed information will be documented in the methodology section of the final report, which is planned in remaining work.

Plan of Remaining Work

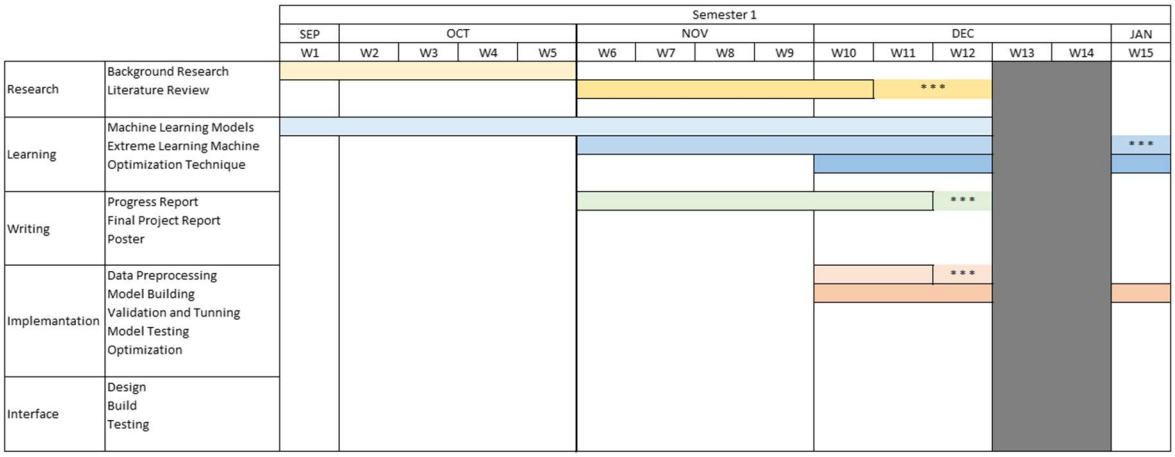

Figure 1. Gantt Chart of Semester 1

As shown in Figure 1, semester 1 was mainly going through learning and research, background research and literature review were done, while machine learning models (Course Module) were learned at the same time. Meanwhile, data preprocessing was performed, and Extreme Learning Machine basics were studied and implemented. The custom optimized model was attempted but still in early stages, while optimization techniques were still in learning progress.

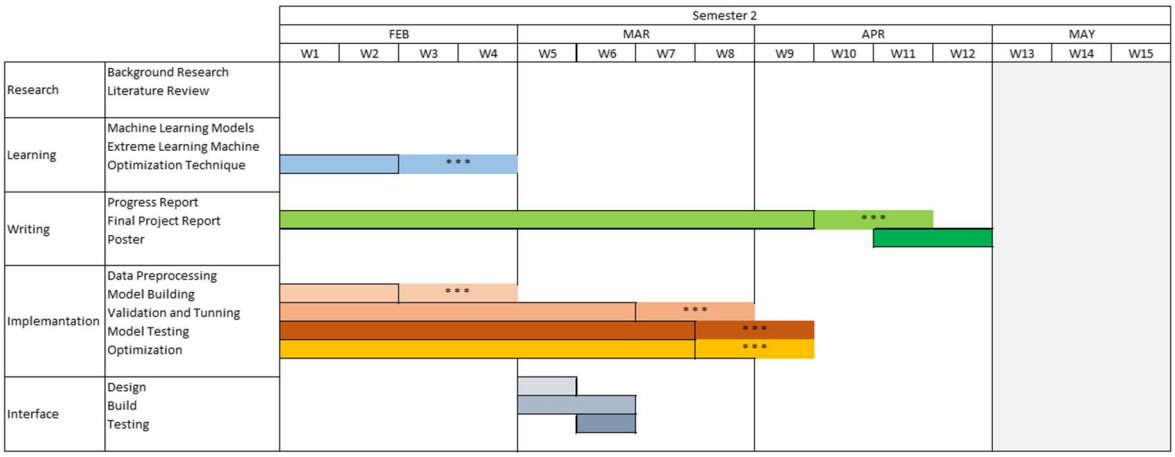

Figure 2. Gantt Chart of Semester 2

The plan for remaining work will be two parts, the remaining weeks of Sem 1 and Sem 2. For the weeks remained in Sem 1, ELM will be studied in advanced by the buffer time, while learning of optimization techniques and building of custom optimized model will be continued. Sem 2 planned as shown in Figure 2, the model building will be continued if not finish, as well as the learning of optimization techniques. Final report writing will be started, along with tunning, testing and optimization of model, which may include inbuilt features selection in the model, apply metaheuristic algorithms such as ABC algorithm. User interface will be built after model is completed while poster will be the last process.

Reference

Ahmed, A.S. *et al.* (2024) “Advancements in Cholelithiasis Diagnosis: A Systematic Review of Machine Learning Applications in Imaging Analysis,” *Cureus* [Preprint]. Available at: https://doi.org/10.7759/cureus.66453.

Akabane, S. *et al.* (2025) “Machine learning-based prediction for incidence of endoscopic retrograde cholangiopancreatography after emergency laparoscopic cholecystectomy: A retrospective, multicenter cohort study,” *Surgical Endoscopy* , 39(3), pp. 1770–1777. Available at: https://doi.org/10.1007/s00464-024-11492-5.

An, Q. *et al.* (2023) “A Comprehensive Review on Machine Learning in Healthcare Industry: Classification, Restrictions, Opportunities and Challenges,” *Sensors*. MDPI. Available at: https://doi.org/10.3390/s23094178.

Ashraf, A.A. *et al.* (2025) “Revolutionizing clinical laboratories: The impact of artificial intelligence in diagnostics and patient care,” *Diagnostic Microbiology and Infectious Disease*. Elsevier Inc. Available at: https://doi.org/10.1016/j.diagmicrobio.2025.116728.

Blum, J. *et al.* (2024) “Using artificial intelligence to predict choledocholithiasis: can machine learning models abate the use of MRCP in patients with biliary dysfunction?,” *ANZ Journal of Surgery*, 94(7–8), pp. 1260–1265. Available at: https://doi.org/10.1111/ans.18950.

Cetin, S. *et al.* (2025) “Determination of Malignancy Risk Factors Using Gallstone Data and Comparing Machine Learning Methods to Predict Malignancy,” *Journal of Clinical Medicine*, 14(17). Available at: https://doi.org/10.3390/jcm14176091.

Esen, I. *et al.* (2024) “Early prediction of gallstone disease with a machine learning-based method from bioimpedance and laboratory data,” *Medicine (United States)*, 103(8), p. E37258. Available at: https://doi.org/10.1097/MD.0000000000037258.

Farnham, B. *et al.* (no date) *Hands-on Machine Learning with Scikit-Learn, Keras, and TensorFlow Concepts, Tools, and Techniques to Build Intelligent Systems SECOND EDITION*.

Götzky, K., Landwehr, P. and Jähne, J. (2013) “Epidemiology and clinical presentation of acute cholecystitis,” *Chirurg*, 84(3), pp. 179–184. Available at: https://doi.org/10.1007/s00104-012-2355-1.

Gutt, C., Schläfer, S. and Lammert, F. (2020) “The treatment of gallstone disease,” *Deutsches Arzteblatt International*, 117(9), pp. 148–158. Available at: https://doi.org/10.3238/arztebl.2020.0148.

Huang, K. *et al.* (2025) “Development and validation of a machine learning-based nomogram for preoperative prediction of laparoscopic surgical difficulty in gallstone patients,” *Translational Gastroenterology and Hepatology*, 10. Available at: https://doi.org/10.21037/tgh-24-124.

Huang, S. *et al.* (2024) *Machine-learning-derived online prediction models of outcomes for patients with cholelithiasis-induced acute cholangitis: development and validation in two retrospective cohorts*. Available at: www.thelancet.com.

Jiang, Y.-G., Ma, X. and Wu, Z. (2026) "Machine learning basics," in *Artificial Intelligence*. Elsevier, pp. 15–50. Available at: https://doi.org/10.1016/b978-0-44-324840-5.00009-3.

Khalifa, M. and Albadawy, M. (2024) "Artificial Intelligence for Clinical Prediction: Exploring Key Domains and Essential Functions," *Computer Methods and Programs in Biomedicine Update*. Elsevier B.V. Available at: https://doi.org/10.1016/j.cmpbup.2024.100148.

Khandelwal, C., Anand, U. and Kant, K. (2025) "Management of Asymptomatic Cholelithiasis," *NMO Journal*, 19(1), pp. 54–60. Available at: https://doi.org/10.4103/jnmo.jnmo_10_25.

Kumar, D. *et al.* (2025) "Computer-aided cholelithiasis diagnosis using explainable convolutional neural network," *Scientific Reports*, 15(1). Available at: https://doi.org/10.1038/s41598-025-85798-2.

Ma, Y. *et al.* (2024) "Early prediction of acute gallstone pancreatitis severity: a novel machine learning model based on CT features and open access online prediction platform," *Annals of Medicine*, 56(1). Available at: https://doi.org/10.1080/07853890.2024.2357354.

Mahalle, P.N. *et al.* (2025) "Fundamentals of machine learning and deep learning," in *Edge Artificial Intelligence*. Elsevier, pp. 15–31. Available at: https://doi.org/10.1016/b978-0-443-26720-8.00008-6.

Mitchell, T.M.. (1997) *Machine Learning*. McGraw-Hill.

Muraviev, S.Y. *et al.* (2025) "Asymptomatic cholecystitis and its controversial issues in biliary surgery," *Asian Journal of Surgery*. Elsevier (Singapore) Pte Ltd, pp. 996–1001. Available at: https://doi.org/10.1016/j.asjsur.2024.10.145.

Pang, S., Ding, T., *et al.* (2019) "A novel YOLOv3-arch model for identifying cholelithiasis and classifying gallstones on CT images," *PLoS ONE*, 14(6). Available at: https://doi.org/10.1371/journal.pone.0217647.

Pang, S., Wang, S., *et al.* (2019) "An artificial intelligent diagnostic system on mobile Android terminals for cholelithiasis by lightweight convolutional neural network," *PLoS ONE*, 14(9). Available at: https://doi.org/10.1371/journal.pone.0221720.

Ruhl, C.E. and Everhart, J.E. (2011) "Gallstone disease is associated with increased mortality in the United States," *Gastroenterology*, 140(2), pp. 508–516. Available at: https://doi.org/10.1053/j.gastro.2010.10.060.

Sakai, Y. *et al.* (2024) "Natural history of asymptomatic gallbladder stones in clinic without beds: A long-term prognosis over 10 years," *World Journal of Clinical Cases*, 12(1), pp. 42–50. Available at: https://doi.org/10.12998/wjcc.v12.i1.42.

Sakorafas, G.H., Milingos, D. and Peros, G. (2007) “Asymptomatic cholelithiasis: Is cholecystectomy really needed? A critical reappraisal 15 years after the introduction of laparoscopic cholecystectomy,” *Digestive Diseases and Sciences*, 52(5), pp. 1313–1325. Available at: https://doi.org/10.1007/s10620-006-9107-3.

Salem, N.M. *et al.* (2023) “Machine and deep learning identified metabolites and clinical features associated with gallstone disease,” *Computer Methods and Programs in Biomedicine Update*, 3. Available at: https://doi.org/10.1016/j.cmpbup.2023.100106.

Sarker, P., Tiang, J.J. and Nahid, A. Al (2025) “Gallstone Classification Using Random Forest Optimized by Sand Cat Swarm Optimization Algorithm with SHAP and DiCE-Based Interpretability,” *Sensors*, 25(17). Available at: https://doi.org/10.3390/s25175489.

Sharma, R., Sharma, S.R. and Prasad, R. (2024) “Gallstone Formation, Diagnosis, Treatment and Prevention,” in R. Sharma, S.R. Sharma, and R. Prasad (eds.) *Gallstone Formation, Diagnosis, Treatment and Prevention*. Academic Press, pp. xv–xvii. Available at: https://doi.org/https://doi.org/10.1016/B978-0-443-16098-1.00021-7.

Srivastava, A. *et al.* (2022) “Automated Prediction of Liver Disease using Machine Learning (ML) Algorithms,” in *2022 2nd International Conference on Advances in Electrical, Computing, Communication and Sustainable Technologies, ICAECT 2022*. Institute of Electrical and Electronics Engineers Inc. Available at: https://doi.org/10.1109/ICAECT54875.2022.9808059.

Stinton, L.M. and Shaffer, E.A. (2012) “Epidemiology of gallbladder disease: Cholelithiasis and cancer,” *Gut and Liver*, pp. 172–187. Available at: https://doi.org/10.5009/gnl.2012.6.2.172.

Unalp-Arida, A. and Ruhl, C.E. (2023) “Increasing gallstone disease prevalence and associations with gallbladder and biliary tract mortality in the US,” *Hepatology*, 77(6), pp. 1882–1895. Available at: https://doi.org/10.1097/HEP.0000000000000264.

Unalp-Arida, A. and Ruhl, C.E. (2024) “Burden of gallstone disease in the United States population: Prepandemic rates and trends,” *World Journal of Gastrointestinal Surgery*, 16(4), pp. 1130–1148. Available at: https://doi.org/10.4240/wjgs.v16.i4.1130.

Wang, X. *et al.* (2024) “Global Epidemiology of Gallstones in the 21st Century: A Systematic Review and Meta-Analysis,” *Clinical Gastroenterology and Hepatology*. W.B. Saunders, pp. 1586–1595. Available at: https://doi.org/10.1016/j.cgh.2024.01.051.

Waring, J., Lindvall, C. and Umeton, R. (2020) “Automated machine learning: Review of the state-of-the-art and opportunities for healthcare,” *Artificial Intelligence in Medicine*. Elsevier B.V. Available at: https://doi.org/10.1016/j.artmed.2020.101822.

Yu, C.J. *et al.* (2021) “Lightweight deep neural networks for cholelithiasis and cholecystitis detection by point-of-care ultrasound,” *Computer Methods and Programs in Biomedicine*, 211. Available at: https://doi.org/10.1016/j.cmpb.2021.106382.